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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Automatic Text Summarization Using a Transformer-Based T5 Model

DSA0306 – Natural Language Processing

Faculty

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ABSTRACT



This project is an **Automatic Text Summarization System** that uses the **Transformer-based T5 model** to generate short and meaningful summaries from long text documents. The system understands the important information in the input text and produces a clear summary while preserving the original meaning. This helps users save time and quickly understand lengthy documents such as articles, reports, and research papers.

Modules:

- Module 1: Advanced Preprocessing and Tokenization
- Module 2: Fine-Tuning T5 for Domain-Specific Summarization
- Module 3: Multi-Document and Long-Text Summarization
- Module 4: Deployment and Real-Time API Integration



INTRODUCTION



- ✓ Text summarization is an important application of Natural Language Processing (NLP).
- ✓ Many documents such as news articles, research papers, and reports contain large amounts of information.
- ✓ Reading the complete document takes time and effort.
- ✓ This project uses the **Transformer-based T5 model** to generate short and meaningful summaries automatically.
- ✓ The generated summary helps users understand the main points quickly without reading the entire document.



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PROBLEM STATEMENT



- Reading long documents takes more time and effort.
- Important information is difficult to identify quickly.
- Manual summarization is time-consuming.

- Automatically generates short and meaningful summaries.
- Identifies the important information from the document.
- Reduces reading time for users.

OBJECTIVES

1

To develop an automatic text summarization system using the T5 Transformer model.

2

To generate short and meaningful summaries from long documents.

3

To reduce the time required to understand large amounts of text.

4

To improve summary quality using the Transformer-based T5 model.



LITERATURE REVIEW



Year	Title of the Paper	Objectives	Limitations
2022	Text Summarization Using Deep Learning	Generate summaries using deep learning models.	Summary quality depends on the dataset.
2023	Transformer-Based Abstractive Summarization	Improve summary quality using Transformer models.	Requires high computational resources.
2024	T5 Model for Automatic Text Summarization	Generate fluent and meaningful summaries using T5.	Performance decreases for very long documents.
2025	Intelligent NLP-Based Text Summarization	Produce accurate summaries using advanced NLP models.	Needs continuous model improvement for better accuracy.



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METHODOLOGY

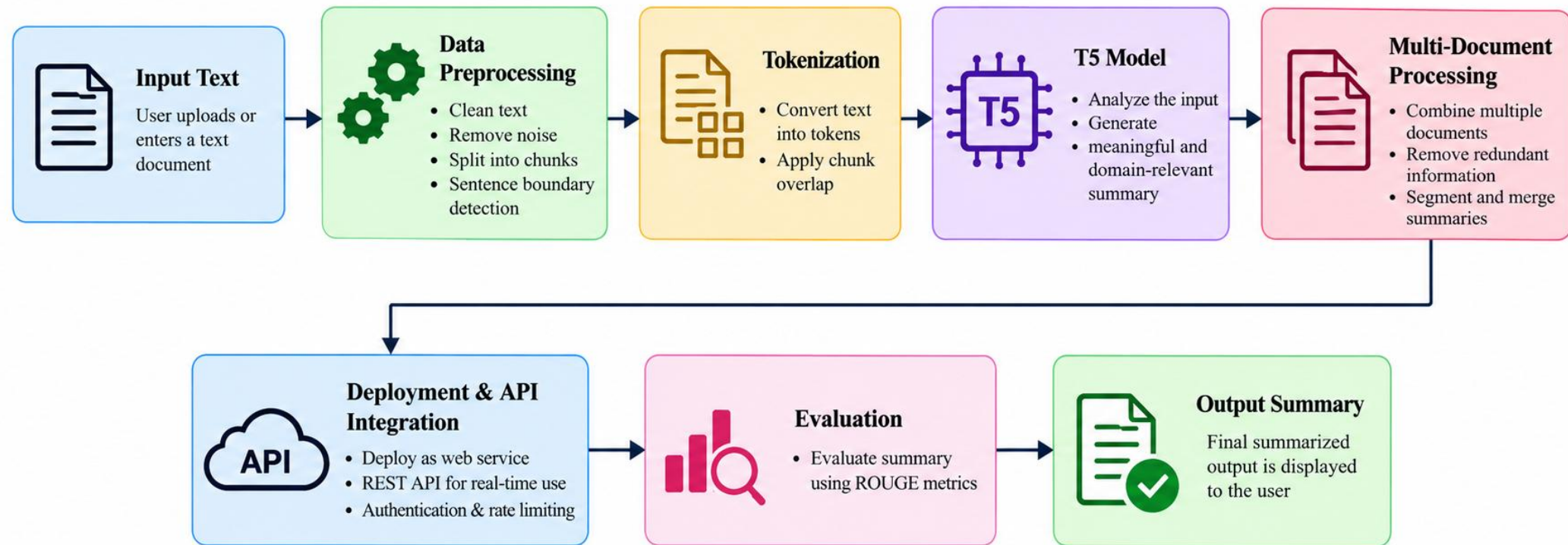


- The user uploads or enters a text document.
 - The system cleans, chunks, and tokenizes the input text.
 - The T5 Transformer model analyzes the document.
 - The model generates a meaningful , domain-relevant summary.
 - The generated summary is evaluated and displayed to the user.
-



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SYSTEM ARCHITECTURE





Module 1:

Advanced Preprocessing and Tokenization



- ✓ Long documents are split into smaller, manageable chunks before processing.
- ✓ Text is cleaned by removing noise, special characters, and extra spaces.
- ✓ Sentence boundaries are detected to preserve meaning during chunking.
- ✓ The T5 tokenizer converts cleaned text into token sequences.
- ✓ Chunk overlap is applied to retain context between adjacent segments.



Module 2:

Fine-Tuning T5 for Domain-Specific Summarization



- ✓ The pretrained T5 model is fine-tuned on domain-specific datasets.
- ✓ Fine-tuning helps the model learn vocabulary specific to the target domain.
- ✓ Hyperparameters such as learning rate and batch size are optimized.
- ✓ Transfer learning reduces the data and time needed for training.
- ✓ The fine-tuned model produces summaries that better match domain needs.



Module 3:

Multi-Document and Long-Text Summarization



- ✓ Multiple related documents are combined into a single input.
- ✓ The system identifies overlapping and redundant information across documents.
- ✓ A unified summary is generated capturing key points from all sources.
- ✓ Documents exceeding the model input limit are summarized in segments.
- ✓ Segment summaries are merged into one coherent final summary.



Module 4:

Deployment and Real-Time API Integration



- ✓ The trained summarization model is packaged and deployed as a web service.
- ✓ A REST API is created to accept text input and return summaries.
- ✓ The API supports real-time requests with minimal response latency.
- ✓ The system is containerized for consistent deployment across environments.
- ✓ Authentication and rate-limiting are added to secure API access.



IMPLIMENTATION



TSSummarizer
NLP ACADEMIC PLATFORM

Dashboard

Summarize

History & Logs

Evaluation

Model Parameters

Domain Fine-Tuning

API Reference

About Project

TS STATUS: READY

Model: TS-v0.0.1

Device: CPU

API KEY CLIENT HEADER

admin123

TRANSFORMER-BASED MODEL

Summarize Long Documents with T5

Transform lengthy articles, PDF reports, and multi-file collections into concise, meaningful summaries using a fine-tuned Transformer-based T5 model.

Start Summarizing

Upload Document

PLATFORM ANALYTICS

FILES SUMMARIZED
11

Stored in local history database

SUMMARIES GENERATED
11

Total T5 inference invocations

AVG COMPRESSION
15%

Percentage of original text size

AVG PROCESSING TIME
7.52s

Latency for T5 chunk calculations

SYSTEM COMPONENTS

Advanced Preprocessing

Handles tokenization, boundary parsing, whitespace clearing, and content sizing before invoking Transformer weights.

Clean Text & Formatting

Sentence boundary detection

Chunking pipeline

Overlap management

T5 Specific tokenizer tokens

Explore Module

Domain Fine-Tuning

Allows uploading datasets of context/targets to optimize the T5 weights for specific legal, medical, or corporate fields.

Domain vocab alignment

T5 pre-training transfer

Learning rate/batch schedules

CSV/JSON datasets

Hyperparameter customization

Explore Module

Long & Multi-Document

Solves the 512 T5 input token limit by chunking, summarizing each, combining, and running a secondary pass to unify content.

Overlapping chunk assembly

Individual sub-summaries

T5 double-pass merging

Redundancy filtration

Multi-file uploads

Explore Module

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Summarization Workspace

Configure advanced settings, process records, and evaluate coverage Quality

Paste raw text

Upload documents

Parameters

Drag and drop files here, or browse files

Supports multiple uploads for academic summarization comparison

Formats: PDF, DOCX, or TXT (Max 10MB per file)

SELECTED DOCUMENTS (1)

NLP_C05_AT3.pdf
125.1 KB

REFERENCE SUMMARY (OPTIONAL FOR ROUGE EVALUATION)

Paste human-written reference summary to calculate precision and recall...

Processing ML Pipeline...

Processing Document Pipeline

Executing natural language processing stages step-by-step

Processing Step 6/7

Reading Document

Extracting text and checking constraints...

Preprocessing text

Removing noise, double spacing, and normalization...

Sentence boundary extraction

Detecting boundaries and parsing structures...

Chunking & Overlap pipeline

Generating overlapping sentence chunks for T5 context window...

TS Tokenizer pass

Converting text chunks into model-specific integer sequences...

TS Model inference

Feeding tensors through standard/fine-tuned Transformer layer weights...

Coherent summary merging

Resolving redundancies and finalizing summaries...

Waiting for Summarization

Input text or upload documents, adjust model params, and trigger 'Generate Summary' to see results.

IMPLIMENTATION

Final summary: 28 words

MODULE 1 PREPROCESSING ANALYSIS

Generated Chunks (6)

Sentences (104)

Normalized Text

Chunk 1	450 words
Chunk 2	399 words
Chunk 3	500 words
Chunk 4	477 words
Chunk 5	448 words
Chunk 6	110 words

CHUNK 1 CONTENT:

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Artificial Intelligence and Data Science ASSESSMENT TOOL 3 - Mock Interview Course Code: DSA03 Course Title: Natural Language Processing CO Assessed: CO5 - Articulation Assessment: Assessment Tool 3 - Mock Interview Weightage: 30% of CIA CO5 Statement: Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5) Student Name: V.Naga Nischay Reg. No.: 192472255 Section / Batch: CSE(AI) Date: 28-08-26 Code of Conduct I V.Naga Nischay (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: Naga Nischay Instructions *

CHUNKER OVERLAP PIPELINE

Summarization History Database

Lightweight SQLite tracking of processed texts, lengths, and latencies

NLP_CO5_AT3.pdf	8/31/2026, 6:27:25 AM	TS: t5-small	Words: 2297 to 28 Time: 35.59s	
Pasted Text	8/31/2026, 3:45:55 PM	TS: t5_finetuned	Words: 10 to 7 Time: 0.66s	
Pasted Text	8/31/2026, 3:45:53 PM	TS: t5-small	Words: 10 to 7 Time: 0.51s	
Pasted Text	8/31/2026, 3:42:34 PM	TS: t5-small	Words: 57 to 28 Time: 3.31s	
Pasted Text	8/31/2026, 3:42:04 PM	TS: t5-small	Words: 57 to 28 Time: 3.74s	
Pasted Text	8/31/2026, 1:17:43 PM	TS: t5-small	Words: 148 to 51 Time: 7.35s	
Pasted Text	8/28/2026, 4:48:13 PM	TS: t5_finetuned	Words: 674 to 62 Time: 14.55s	
Pasted Text	8/28/2026, 4:30:41 PM	TS: t5-small	Words: 56 to 28 Time: 5.45s	
Pasted Text	8/28/2026, 4:28:26 PM	TS: t5-small	Words: 127 to 25 Time: 17.61s	



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RESULTS & DISCUSSION



The fine-tuned T5 model produced more accurate and domain-relevant summaries than the base model. ROUGE scores showed a noticeable improvement in summary quality after fine-tuning on domain-specific data.

Multi-document summarization successfully reduced redundant information across related sources. The advanced preprocessing module ensured longer documents were handled without losing important context.

The deployed API delivered summaries within a response time suitable for real-time use. Overall results confirm that the four new modules meaningfully improved the system's capability and usability.



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CONCLUSION



The project successfully extends automatic text summarization with four new modules: advanced preprocessing, domain fine-tuning, multi-document summarization, and real-time API deployment.



Fine-tuning and improved preprocessing enabled the T5 model to generate more accurate, domain-appropriate summaries from longer and more complex documents.



Real-time API deployment made the system practical for real-world use, making the summarization pipeline more accurate, scalable, and usable overall.